

TECHNICAL REPORT

What Drives AI Job Salaries?

An Exploratory Analysis of Compensation in the Global AI Job Market

A Descriptive and Visual Analysis of 15,000 AI Job Postings Worldwide

Tools: Python · pandas · NumPy · Matplotlib · Seaborn

Notebook: [aijobs.ipynb](#)

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1. Introduction

1.1 Background

Artificial intelligence has become one of the fastest-growing employment categories in the global technology sector, spanning roles from research scientists to MLOps engineers to AI consultants. As demand for AI talent has grown, so has variation in how that talent is compensated — salaries differ sharply by seniority, required skill set, company size, geography, and job structure. Understanding which of these factors actually drives compensation, as opposed to which merely correlate with it by coincidence, is valuable to job seekers negotiating an offer, to employers setting competitive pay bands, and to analysts tracking the shape of an emerging labour market.

This project uses a dataset of 15,000 AI job postings to explore a practical question: **which factors are most strongly associated with AI salaries, and which commonly assumed drivers of pay — such as formal education or remote-work arrangement — turn out to matter less than expected?**

1.2 Why This Project Matters

This project has value on three levels.

- **Practically**, distinguishing genuine salary drivers from weak or coincidental ones is exactly the kind of groundwork a compensation analyst, recruiter, or job seeker needs before making a pay-related decision — for example, discovering that education level shows almost no salary premium in this dataset is directly useful information for someone deciding whether an additional credential is worth pursuing purely for pay.
- **Analytically**, this project is deliberately different in kind from the two regression projects that preceded it. It is a descriptive, exploratory analysis rather than a predictive modelling exercise — no target variable is predicted, and no train/test split or machine learning model is built. Carrying out a rigorous, well-structured exploratory analysis is its own distinct skill from building a regression model, and this project was chosen specifically to practice and demonstrate that skill on its own terms.
- **Developmentally**, the base code for this analysis was adapted from a public Kaggle notebook by Omar Yasser, with modifications made during adaptation. This technical report — including all interpretation, cross-referencing between charts, and the analytical narrative connecting the findings — was written independently. Building a report of this depth on top of borrowed code is itself a distinct exercise: it requires understanding what the code actually does and does not show, well enough to explain it accurately and critique it honestly, rather than simply running it.

1.3 Objectives

The project set out to:

1. Load and inspect a large, real-world AI job postings dataset covering compensation, seniority, skills, and company attributes.
2. Confirm data quality — checking for missing values, duplicate records, and correct data types — before drawing any conclusions.
3. Explore the distribution of salary and every relevant categorical and numerical factor through a structured set of visualisations.
4. Identify which factors show a strong, visible relationship with salary, and which do not, using both statistical correlation and targeted charts.
5. Extract and rank the most in-demand technical skills across all 15,000 postings.
6. Translate the findings into practical observations for job seekers, employers, and future analysis.

1.4 Scope and Approach

Unlike the two regression projects that preceded this one, this analysis does **not** build a predictive model. There is no target/feature split, no scaling, no train/test split, and no machine learning algorithm fitted. The scope here is **exploratory and descriptive**: using summary statistics, distribution plots, grouped comparisons, and correlation analysis to understand the structure of the data and surface genuine patterns, without attempting to predict an unseen salary value. This distinction is stated explicitly up front because it changes what "preprocessing" means in Section 3 — there is no feature engineering pipeline to build, only data-quality checks and light preparation for specific charts.

1.5 Tools and Libraries

Library	Role in this project
pandas	Loading the CSV, inspecting structure, datetime conversion, and all DataFrame manipulation
NumPy	Underlying numerical array operations
Matplotlib	Figure sizing and layout for all charts
Seaborn	Histograms, bar charts, boxplots, violin plots, count plots, scatter plot, heatmap, and pairplot

1.6 A Note on Code Origin and Authorship

The base analytical code used in this notebook was adapted from a publicly available Kaggle notebook authored by Omar Yasser, with modifications made during adaptation for this specific report. This technical report itself — the structure, the written interpretation of every chart and statistic, the cross-referencing

between findings, and the recommendations — was researched and written independently. Section 10.2 provides full attribution for the original code source.

2. Story of the Data

2.1 Source and Structure

The dataset, `ai_job_dataset.csv`, contains **15,000 records and 19 columns**. Each row represents a single AI-related job posting, combining compensation data, role and seniority information, required skills, employer characteristics, and posting-level metadata such as description length and application deadline.

Confirmed via `df.info()`:

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 15000 entries, 0 to 14999  
Data columns (total 19 columns):  
dtypes: float64(1), int64(4), object(14)  
memory usage: 2.2+ MB
```

2.2 Data Dictionary

#	Column	Description
0	job_id	Unique identifier for the posting (e.g. AI00001)
1	job_title	Job title (e.g. AI Research Scientist, NLP Engineer)
2	salary_usd	Annual salary, converted to USD — the primary variable of interest
3	salary_currency	Original currency of the salary offer (USD, EUR, or GBP)
4	experience_level	Seniority: EN (Entry), MI (Mid), SE (Senior), EX (Executive)
5	employment_type	CT (Contract), FL (Freelance), FT (Full-time), PT (Part-time)
6	company_location	Country where the hiring company is based
7	company_size	S (Small), M (Medium), L (Large)
8	employee_residence	Country where the employee resides
9	remote_ratio	Percentage of work performed remotely: 0, 50, or 100
10	required_skills	Comma-separated list of skills required for the role
11	education_required	Minimum education level: Associate, Bachelor, Master, PhD
12	years_experience	Years of professional experience (0–19)
13	industry	Industry sector of the hiring company
14	posting_date	Date the job was posted

#	Column	Description
15	application_deadline	Deadline for applications
16	job_description_length	Character length of the job description
17	benefits_score	A numeric score (5–10) summarising the generosity of benefits offered
18	company_name	Name of the hiring company

2.3 Data Quality

The dataset arrived in excellent condition. `df.isna().sum()` returned zero missing values across all 19 columns, and `df.duplicated().sum()` returned zero duplicate rows. This meant no imputation, deduplication, or missing-value strategy was needed — a contrast with the earlier student-performance project, where 9.1% of one column required imputation.

One light data-type correction was applied: `posting_date` and `application_deadline` arrived as plain text and were converted to proper datetime objects.

```
df['posting_date'] = pd.to_datetime(df['posting_date'])
df['application_deadline'] = pd.to_datetime(df['application_deadline'])
```

This conversion does not change any values, but it allows pandas to treat these columns as genuine dates rather than arbitrary text — relevant for any future work involving posting duration or seasonal trends, though no such analysis was performed in this notebook.

2.4 Summary Statistics

Statistic	salary_usd	years_experience	benefits_score
count	15,000	15,000	15,000
mean	115,348.97	6.25	7.50
std	60,260.94	5.55	1.45
min	32,519	0	5.00
25%	70,179.75	2	6.20
50%	99,705	5	7.50
75%	146,408.50	10	8.80
max	399,095	19	10.00

The wide gap between the mean (\$115,349) and the maximum (\$399,095), combined with a standard deviation (\$60,261) that is more than half the mean, is an early signal that `salary_usd` is right-skewed rather than symmetric — confirmed visually in Section 4.

2.5 A Notable Structural Pattern in the Data

One observation is worth flagging honestly before any further analysis: the relationship between `years_experience` and `experience_level` appears to be deterministic rather than naturally varied. As Figure 12 (Section 7) shows, every Entry-level (EN) posting falls at exactly 0–1 years of experience, every Mid-level (MI) posting at exactly 2–4 years, every Senior (SE) posting at exactly 5–9 years, and every Executive (EX) posting at exactly 10–19 years, with no overlap or exceptions at all between bands.

In genuine real-world hiring data, these bands would be expected to overlap considerably — a real dataset would very likely include, for example, an entry-level hire with two years of prior experience, or a senior professional with only four years in the field. The complete absence of any overlap here is a strong indicator that *years_experience was very likely generated from experience_level (or vice versa) using a fixed rule, rather than collected as an independently observed value* — which does not reduce the value of the exploratory patterns found (salary differences by seniority are still worth understanding), but it does mean that any variable directly derived from this banding structure should not be over-interpreted as reflecting the full complexity of real-world career progression.

3. Data Preparation and Preprocessing

Because this project is exploratory rather than predictive, this section differs structurally from a typical machine-learning preprocessing stage. There is no target/feature split, no categorical encoding for a model, no feature scaling, and no train/test split — none of these are meaningful without a model being trained. What follows is the preparation that was actually needed to support accurate description and visualisation of the data.

3.1 Datetime Conversion

```
df['posting_date'] = pd.to_datetime(df['posting_date'])
df['application_deadline'] = pd.to_datetime(df['application_deadline'])
```

As noted in Section 2.3, these two columns were converted from text to proper datetime objects, confirmed by a second `df.info()` call showing their dtype changed to `datetime64[ns]`.

3.2 Parsing the Skills Column

`required_skills` arrives as a single comma-separated string per row (e.g. "Tableau, PyTorch, Kubernetes, Linux, NLP"). To analyse individual skills rather than whole strings, this column was split and cleaned:

```
df["skills_list"] = df["required_skills"].apply(
    lambda x: [s.strip() for s in x.split(",")]
)
```

The lambda function splits each string on commas and strips leading/trailing whitespace from each resulting skill name, producing a Python list per row (e.g. ['Tableau', 'PyTorch', 'Kubernetes', 'Linux', 'NLP']). This list-per-row structure is what makes the skill-frequency analysis in Section 3.3 possible.

3.3 Flattening and Counting Skills

```
all_skills = [skill for sub in df["skills_list"] for skill in sub]
skills_counts = pd.Series(all_skills).value_counts()
```

The nested list comprehension flattens every row's list of skills into one long list covering all 15,000 postings combined (each posting typically lists 4–5 skills, so this list runs into the tens of thousands of entries). Wrapping it in `pd.Series(...).value_counts()` then counts how many postings mention each individual skill, which is what powers the "Top 20 Skills" chart in Section 7.

3.4 Selecting Numeric Columns for Correlation

```
numeric_cols = ["salary_usd", "remote_ratio"]  
if "benefits_score" in df.columns:  
    numeric_cols.append("benefits_score")
```

A short list of genuinely continuous numeric columns was assembled for the correlation heatmap and pairplot in Section 5. The conditional check for `benefits_score` is a defensive habit — useful if this code were reused on a version of the dataset that lacked that column — though in this dataset the column is always present.

Note: `years_experience`, despite being numeric, was not included in this correlation list in the original notebook. Given the deterministic relationship with `experience_level` documented in Section 2.5, and its clear visual relationship with salary in Figure 12, its omission here is worth revisiting — see Section 8.3.

4. Pre-Analysis

Before drawing conclusions, the shape of the central variable — salary — and its surrounding categorical context were examined on their own terms.

4.1 Salary Distribution

```
sns.histplot(df['salary_usd'], kde=True, color="purple", bins=50, edgecolor="black")
```

The salary distribution (Figure 1, Section 7) is clearly **right-skewed**: the bulk of postings cluster between roughly \$50,000 and \$150,000, with a long thinning tail extending out to the maximum of \$399,095. This shape is typical of income and compensation data generally — most roles cluster at a common range, while a smaller number of senior or executive positions pull the tail far to the right.

4.2 Currency Distribution

The vast majority of postings are denominated in USD, with EUR and GBP each representing a much smaller share (Figure 2, Section 7). It is worth noting that salary_usd already reflects each posting's value converted to a common USD basis regardless of its original currency, which is why comparisons across the whole dataset in later sections remain valid despite this currency mix.

4.3 Categorical Variable Overview

Experience level, education required, company size, and remote ratio were each examined individually before any cross-tabulation against salary:

- `experience_level` has four categories (EN, MI, SE, EX) as described in the data dictionary, with a healthy number of postings in each band.
- `education_required` spans four levels (Associate, Bachelor, Master, PhD), with Bachelor and Master the most commonly required (Figure 13, Section 7).
- `company_size` has three categories (S, M, L), each represented in meaningful numbers (Figure 11, Section 7).
- `remote_ratio` takes exactly three values — 0 (fully on-site), 50 (hybrid), and 100 (fully remote) — fairly evenly distributed across the dataset (Figure 10, Section 7).

5. In-Analysis

5.1 Salary by Experience Level

```
sns.barplot(x="experience_level",  
y="salary_usd", data=df, color="purple")
```

Experience Level	Approx. Average Salary (USD)
EN (Entry)	≈ \$63,000
MI (Mid)	≈ \$88,000
SE (Senior)	≈ \$122,000
EX (Executive)	≈ \$188,000

The pattern is clean and monotonic: average salary rises at every step up the seniority ladder, roughly tripling from Entry to Executive level. This is the single clearest, most consistent relationship in the entire dataset, and it reappears in nearly every other chart that includes experience level as a grouping variable (Figures 5, 6, and 12).

5.2 Top Skills in Demand

Ranking all mentioned skills by frequency across the 15,000 postings (Figure 4, Section 7) produced a clear ordering, led by:

Rank	Skill	Approx. Postings Mentioning It
1	Python	≈ 4,450
2	SQL	≈ 3,400
3	TensorFlow	≈ 3,000
4	Kubernetes	≈ 3,000
5	Scala	≈ 2,800
6	PyTorch	≈ 2,750

Python leads by a clear margin, appearing in well over a quarter of all postings — a similar dominance to what "study hours" and "smoking status" showed in the two earlier regression projects, in the sense that one factor stands out clearly ahead of the rest. The remainder of the top 20 (visible in full in Figure 4) is a mix of programming languages (SQL, Scala, Java, R), cloud platforms (GCP, Azure), ML frameworks (TensorFlow, PyTorch), and infrastructure tools (Kubernetes, Docker, Linux, Git) — reflecting the genuinely cross-disciplinary nature of AI roles, which blend software engineering, data science, and MLOps skill sets.

5.3 Correlation Analysis — A Genuinely Surprising Result

```
corr =  
df[numeric_cols].corr()  
sns.heatmap(corr, annot=True,  
cmap="coolwarm")
```

Variable Pair	Correlation	Interpretation
salary_usd – remote_ratio	+0.014	Essentially zero — no linear relationship
salary_usd – benefits_score	+0.001	Essentially zero — no linear relationship
remote_ratio – benefits_score	+0.003	Essentially zero — no linear relationship

Every single correlation in this matrix rounds to zero. Neither how remote a role is, nor how generous its benefits package is, shows any meaningful linear relationship with salary in this dataset. This is a genuinely useful negative finding, not a failed analysis — it tells us clearly that **salary in this dataset is not being set as a trade-off against remote flexibility or benefits generosity**; whatever is driving compensation, it is not these two numeric factors.

This result is echoed directly in the boxplots in Section 7: Figure 8 (salary by remote ratio) shows three nearly identical box shapes, and there is no equivalent chart for benefits_score against salary in this notebook — a natural addition suggested in Section 8.3.

5.4 Why the Real Signal Is Categorical, Not Numeric

Taken together, Sections 5.1 and 5.3 point to an important structural conclusion: of the columns examined numerically, only years_experience (via its proxy, experience_level) shows a strong relationship with salary. The two numeric columns actually included in the correlation matrix (remote_ratio and benefits_score) show none. This means the real explanatory power in this dataset lives almost entirely in categorical and ordinal groupings — seniority level, and to a lesser extent job title, industry, and company size — rather than in the small set of purely numeric columns available.

6. Post-Analysis and Insights

6.1 Seniority Is the Dominant, Consistent Driver of Salary

Every chart that cross-references experience level against salary — the bar chart (Figure 3), the violin plot (Figure 5), the boxplot (Figure 6), and the scatter plot against years of experience (Figure 12) — tells the same consistent story: salary rises steadily and substantially with seniority, from roughly \$63,000 at Entry level to roughly \$188,000 at Executive level. This consistency across multiple independent chart types, rather than a single statistic, is what makes this the most trustworthy finding in the report.

6.2 Education Level Shows Almost No Salary Premium — A Genuinely Counter-Intuitive Finding

This is the most surprising result in the entire analysis. The boxplot of salary by education level (Figure 7) shows four nearly identical distributions — Associate, Bachelor, Master, and PhD postings all have almost the same median salary (roughly \$97,000–\$103,000) and almost the same spread. **A PhD does not command a visibly higher salary than a Bachelor's degree in this dataset.** This runs counter to the common assumption that more formal education reliably translates into higher pay, and it is a finding worth stating plainly rather than glossing over: at least in this dataset, seniority and demonstrated experience appear to matter far more than credential level.

6.3 Remote Work Arrangement Has No Measurable Effect on Pay

Section 5.3's near-zero correlation (+0.014) between `remote_ratio` and salary is directly confirmed visually: the boxplot in Figure 8 shows on-site (0), hybrid (50), and fully remote (100) roles all clustering around the same median and interquartile range. In this dataset, working remotely is neither a pay premium nor a pay penalty — it appears to be a logistical arrangement independent of compensation.

6.4 Company Size Has a Real but Modest Effect

Unlike remote ratio and education, company size does show a visible, if modest, gradient (Figure 9): median salary is highest at Large companies, in the middle at Medium companies, and lowest at Small companies. The effect is real but far smaller than the seniority effect — the difference between company sizes spans perhaps \$25,000 at the median, compared to roughly \$125,000 between Entry and Executive seniority.

6.5 Python's Dominance Reflects the Cross-Disciplinary Nature of AI Roles

Python's clear lead in Section 5.2's skill ranking, ahead of SQL, TensorFlow, and Kubernetes, is consistent with its well-documented role as the common language bridging data science, machine learning engineering, and MLOps. The presence of both ML-specific tools (TensorFlow, PyTorch) and general

infrastructure tools (Kubernetes, Docker, Linux) in the top 10 confirms that AI job postings, as a category, blend software engineering and data science skill sets rather than sitting cleanly in either.

6.6 A Necessary Caveat on Data Realism

The structural observation in Section 2.5 — that `years_experience` bands map perfectly and without exception onto `experience_level` categories — should inform how confidently these findings are generalised to the real-world AI job market. This pattern strongly suggests the dataset was at least partly synthetically generated or algorithmically labelled, rather than being organically scraped from real job postings with all their natural noise and overlap. This does not invalidate the patterns found — the salary-by-seniority relationship, the skill rankings, and the near-zero effect of remote ratio are all genuine features of the data as given — but it does mean the findings should be understood as descriptive of this specific dataset's structure, rather than assumed to transfer perfectly to messier real-world hiring data without further validation.

7. Data Visualisations and Charts

7.1 Figure 1 — Distribution of AI Salaries

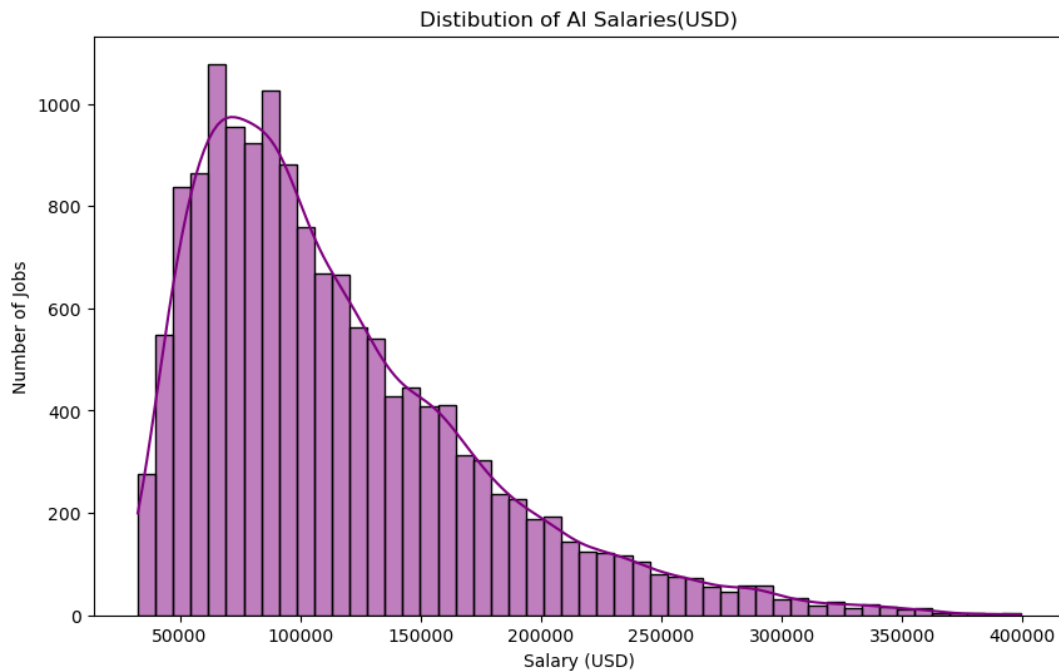


Figure 1. Distribution of salary_usd across all 15,000 postings, with a kernel density curve overlaid.

What it shows: a clear right-skew, with most salaries clustering between \$50,000 and \$150,000 and a long tail extending toward \$400,000.

7.2 Figure 2 — Number of Jobs by Salary Currency

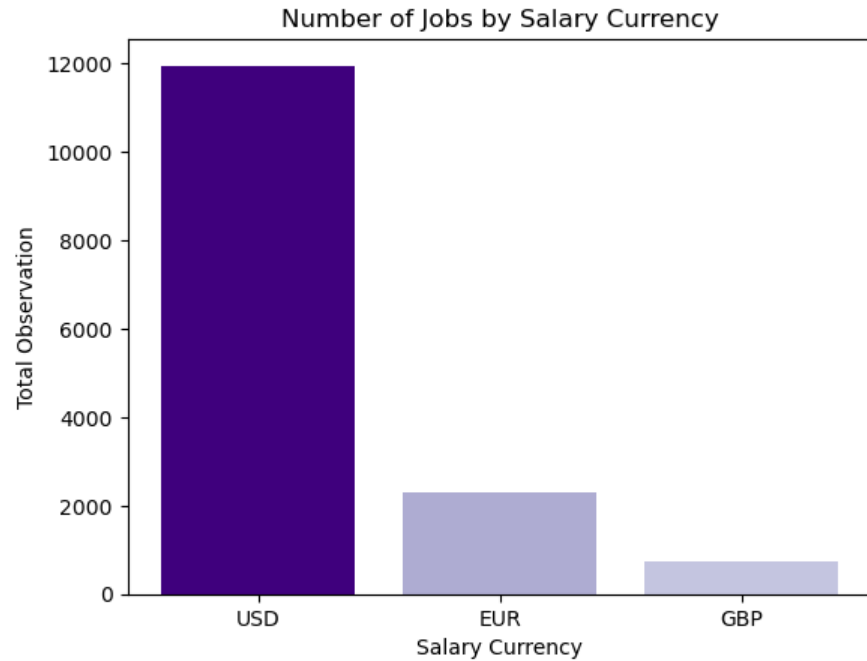


Figure 2. Count of postings by original salary currency.

What it shows: USD dominates the postings by a wide margin, with EUR and GBP each contributing a much smaller share.

7.3 Figure 3 — Average Salary by Experience Level



Figure 3. Mean salary_usd for each experience_level category.

What it shows: a clean, monotonic rise in average salary from Entry through Executive level, the strongest pattern in the dataset.

7.4 Figure 4 — Top 20 Skills in AI Jobs

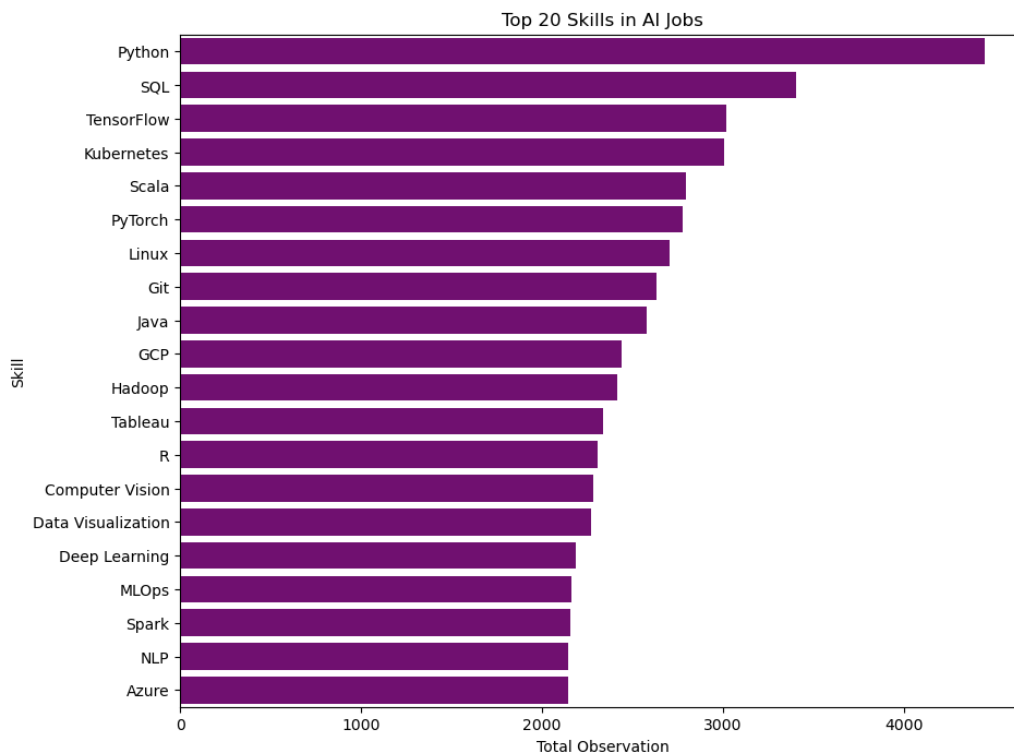


Figure 4. The 20 most frequently required skills across all postings, by count.

What it shows: Python leads clearly, followed by SQL, TensorFlow, and Kubernetes, with a long, fairly even tail of specialised tools.

7.5 Figure 5 — Salary Distribution by Experience Level (Violin Plot)

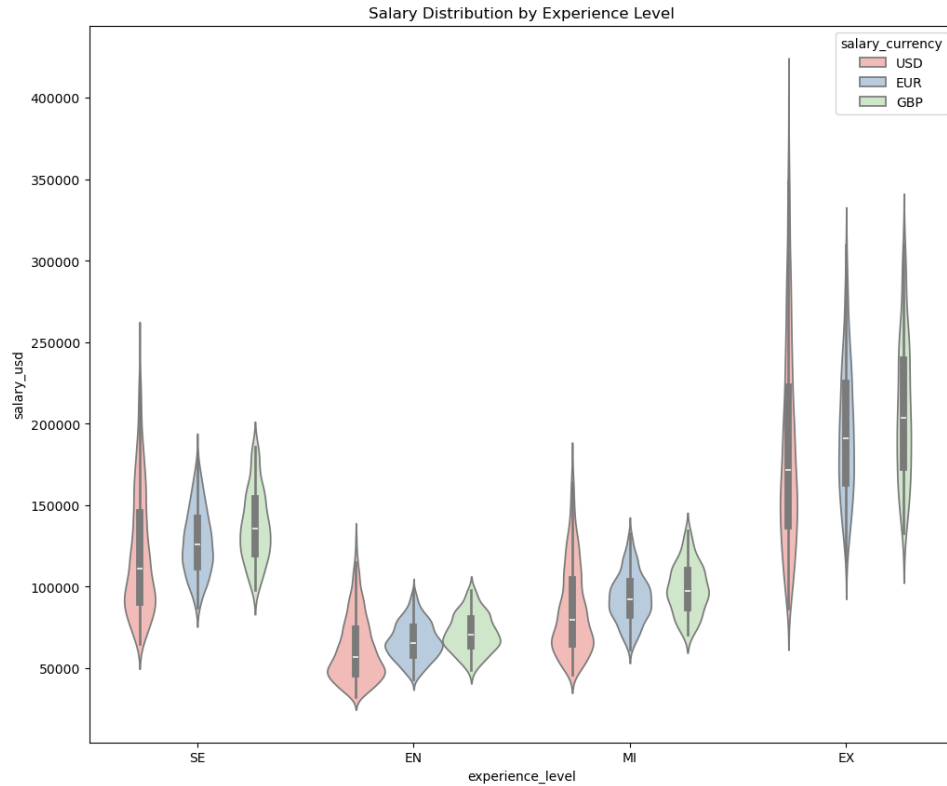


Figure 5. Full salary distribution shape for each experience level, split by salary currency.

What it shows: not just the average but the full spread of salaries within each seniority band, confirming that the upward shift with seniority holds across the whole distribution, not just at the mean.

7.6 Figure 6 — Salary by Experience Level (Boxplot)

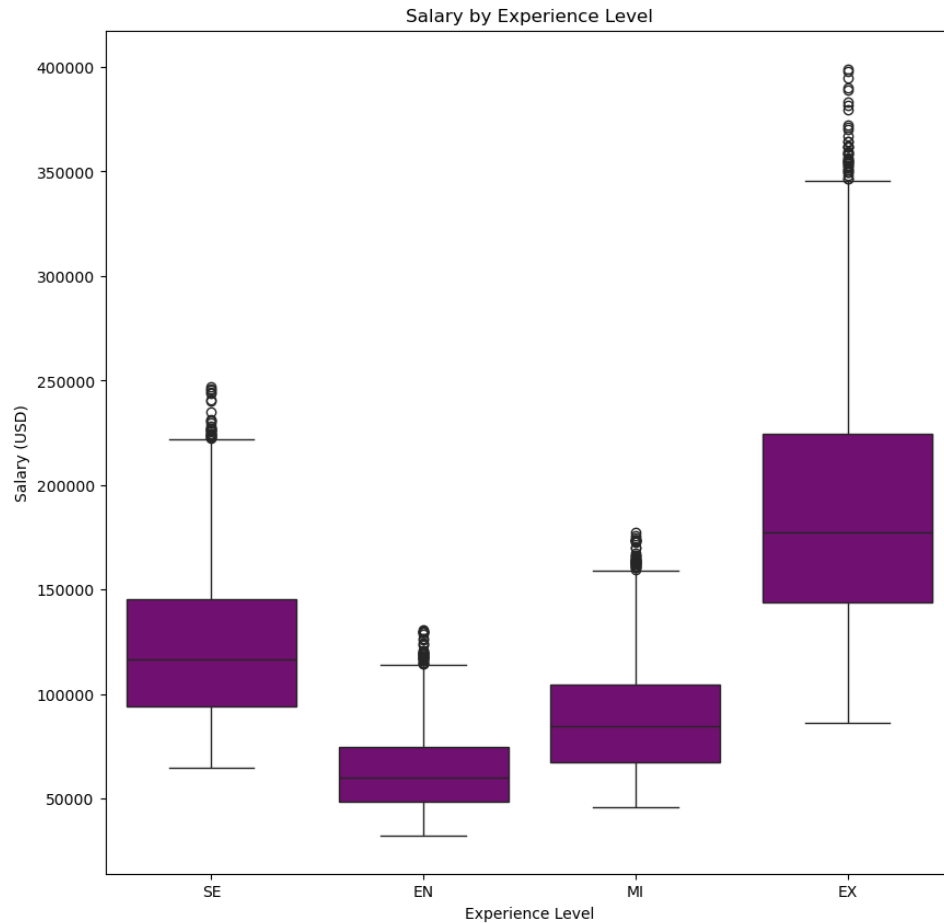


Figure 6. Median, quartiles, and outliers of salary_usd for each experience level.

What it shows: a clean, non-overlapping upward step in median salary from EN to EX, the sharpest categorical divide of any chart in the report.

7.7 Figure 7 — Salary by Education Level (Boxplot)

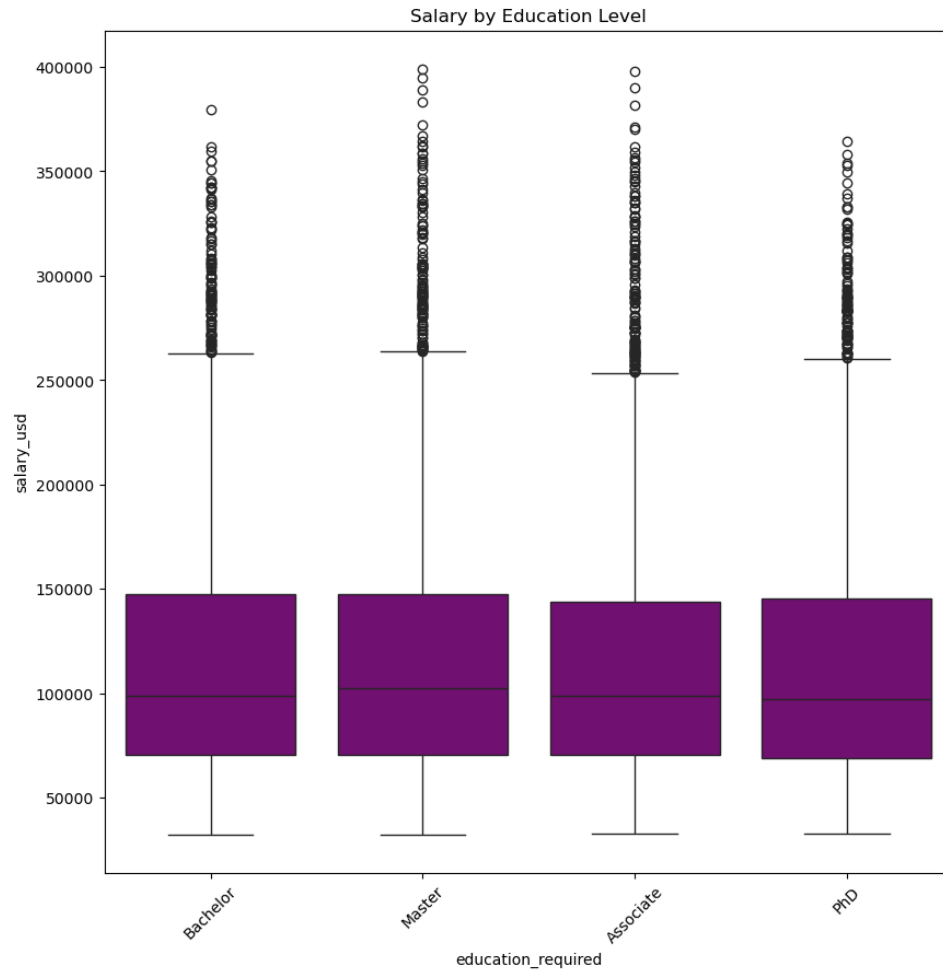


Figure 7. Median, quartiles, and outliers of salary_usd for each required education level.

What it shows: four nearly identical boxes — the counter-intuitive finding that formal education level shows almost no relationship with salary in this dataset.

7.8 Figure 8 — Salary vs. Remote Ratio (Boxplot)

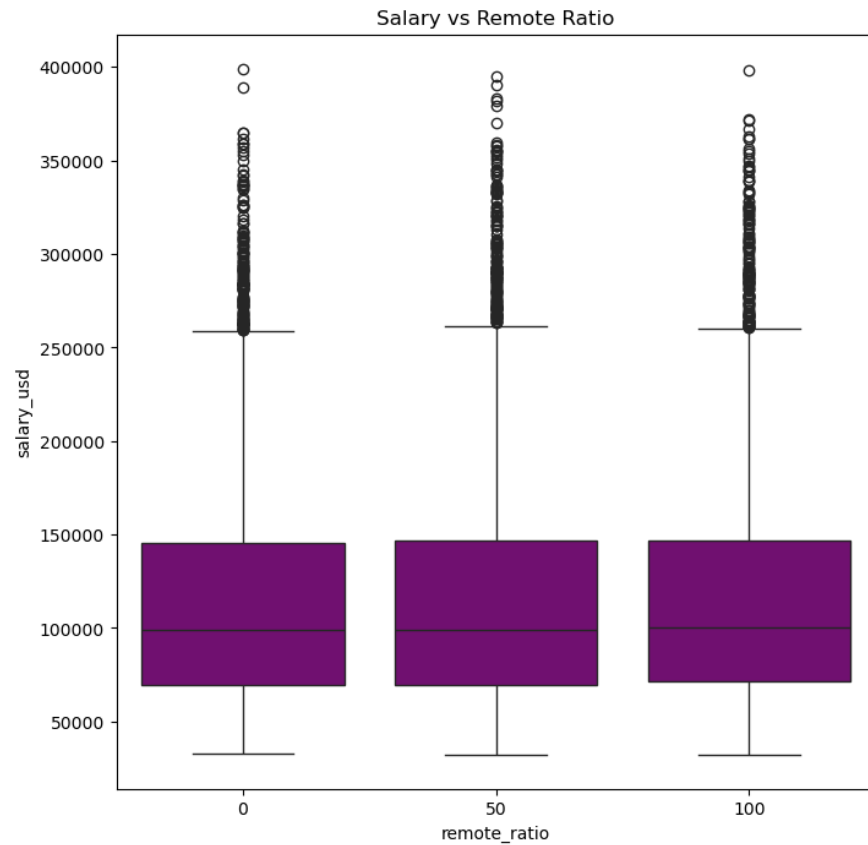


Figure 8. Salary distribution at 0%, 50%, and 100% remote work.

What it shows: three virtually indistinguishable distributions, confirming the near-zero correlation found in Section 5.3.

7.9 Figure 9 — Salary by Company Size (Boxplot)

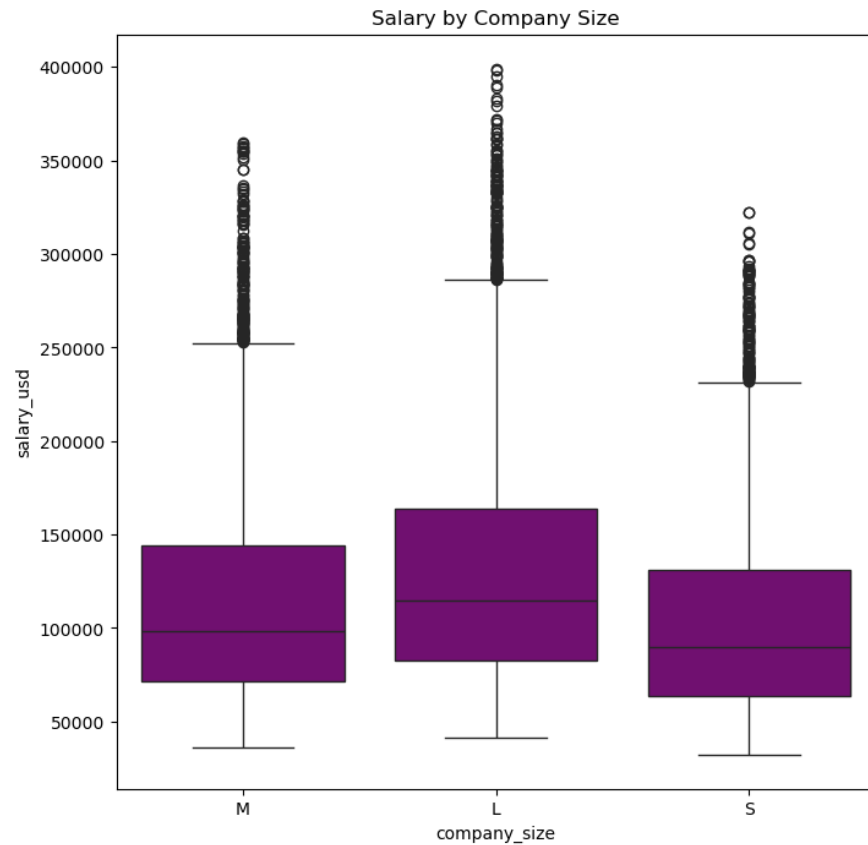


Figure 9. Salary distribution for Small, Medium, and Large companies.

What it shows: a modest but real gradient, with Large companies paying the most and Small companies the least at the median.

7.10 Figure 10 — Remote Ratio Distribution

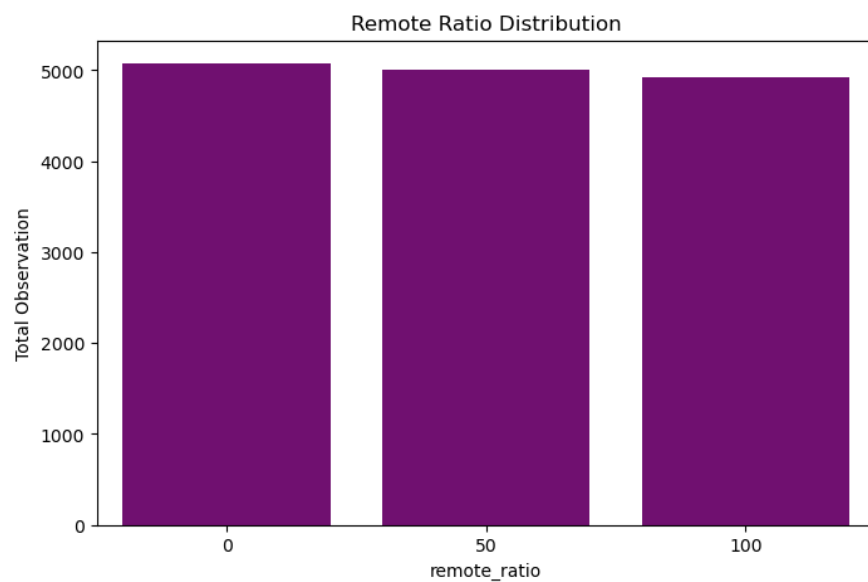


Figure 10. Count of postings at each remote-work level.

What it shows: a reasonably even split of postings across on-site, hybrid, and fully remote arrangements.

7.11 Figure 11 — Total Observations by Company Size

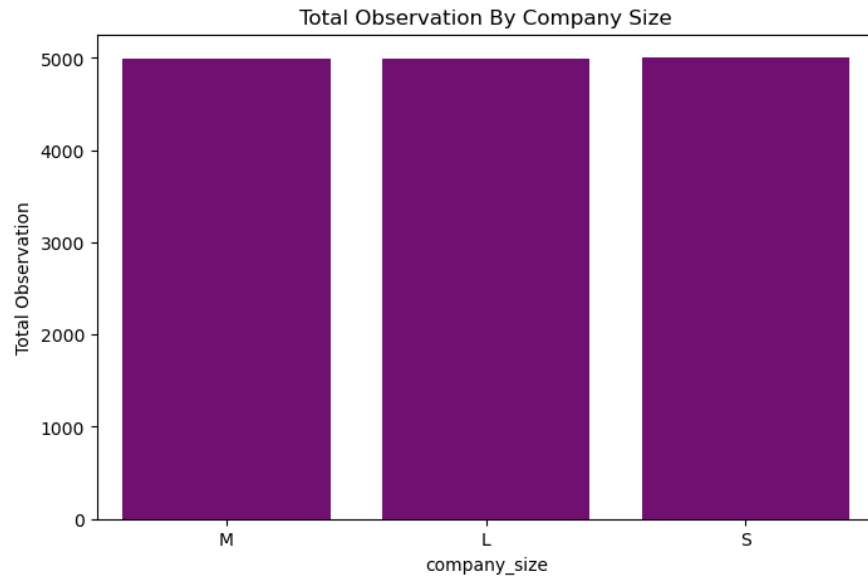


Figure 11. Count of postings by company size category.

What it shows: all three company-size categories are meaningfully represented in the dataset, supporting the comparison in Figure 9.

7.12 Figure 12 — Salary vs. Years of Experience



Figure 12. Every posting plotted by years of experience and salary, coloured by experience level.

What it shows: the structural pattern discussed in Sections 2.5 and 6.6 — each experience-level colour occupies a completely distinct, non-overlapping band of years_experience, with salary rising within and across those bands.

7.13 Figure 13 — Education Required for AI Jobs

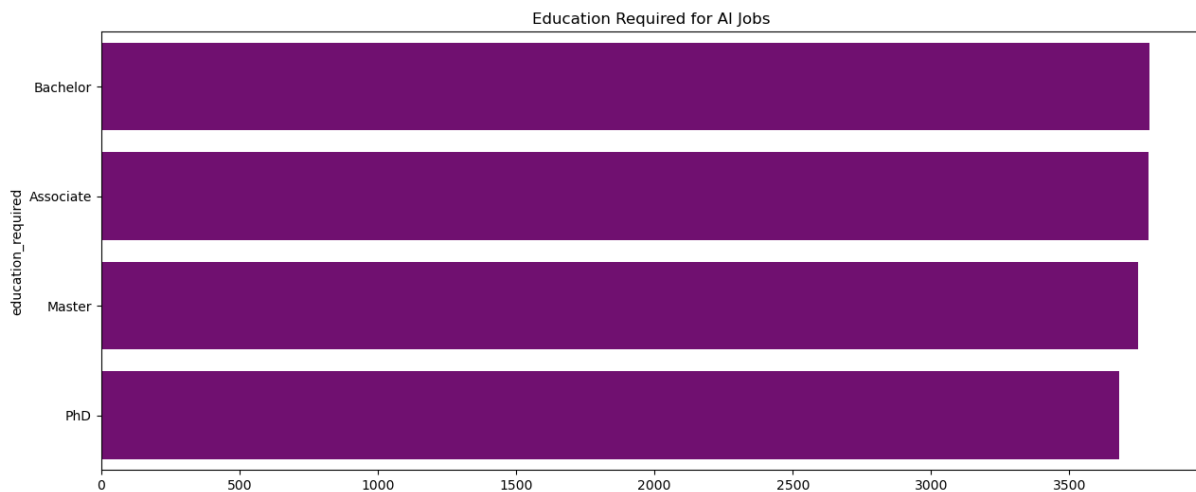


Figure 13. Count of postings by required education level.

What it shows: Bachelor's and Master's degrees are the most commonly required credentials, with Associate and PhD requirements less frequent.

7.14 Figure 14 — Correlation Matrix

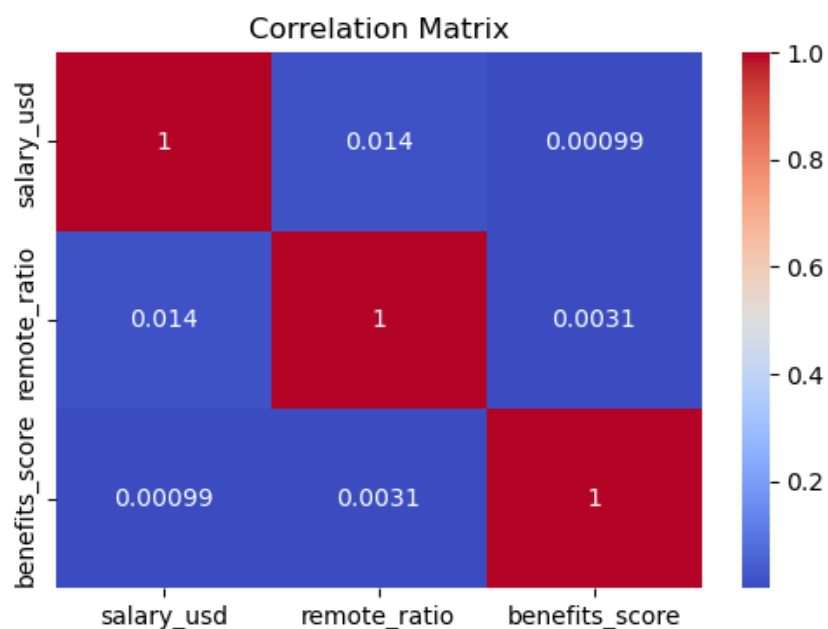


Figure 14. Correlation between salary_usd, remote_ratio, and benefits_score.

What it shows: every off-diagonal value rounds to essentially zero — visual confirmation that neither remote ratio nor benefits score has a meaningful linear relationship with salary.

7.15 Figure 15 — Pairplot

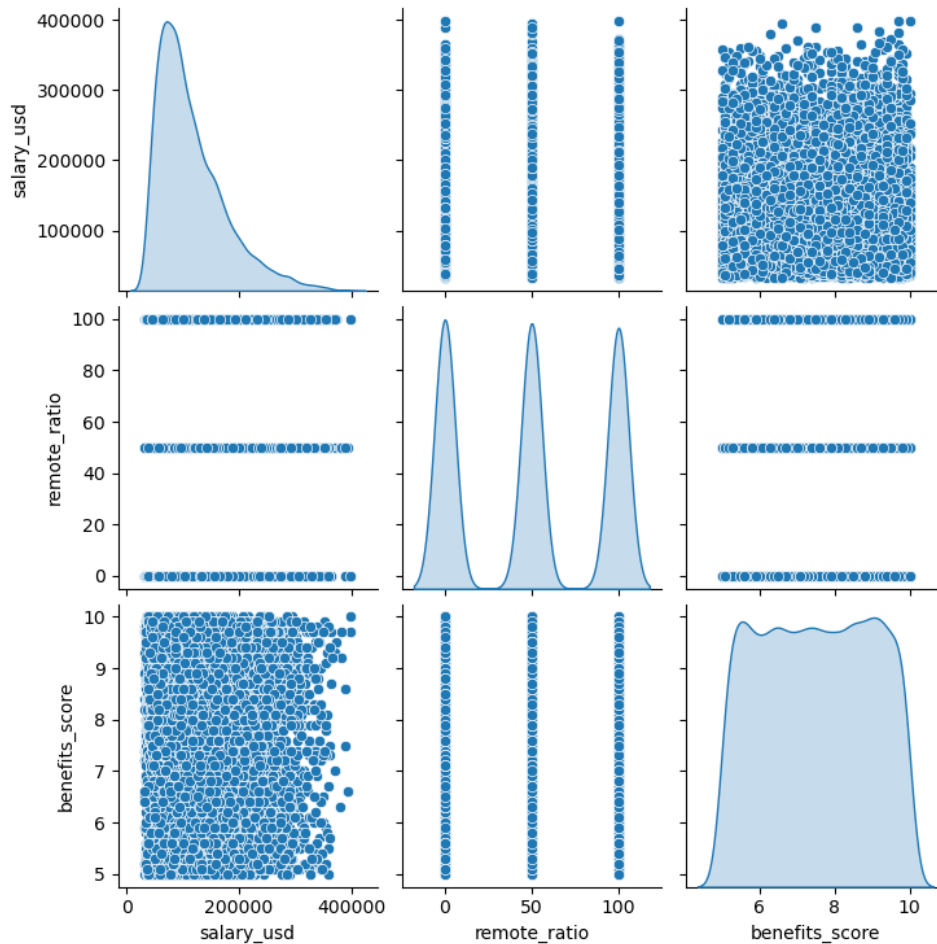


Figure 15. Pairwise scatter relationships and individual distributions for salary_usd, remote_ratio, and benefits_score.

What it shows: a consolidated view confirming Figure 14 — no visible pairwise structure between the three numeric variables beyond salary’s own right-skewed shape on the diagonal.

8. Recommendations and Observations

8.1 Recommendations for Job Seekers and Employers

1. Prioritise demonstrated seniority and experience over additional formal credentials when the goal is salary growth.

With education level showing almost no measurable salary premium in this dataset, a job seeker weighing an additional degree purely for compensation reasons should weigh that decision carefully against gaining additional years of hands-on, senior-level experience instead.

2. Treat Python, SQL, and cloud/MLOps tooling as the baseline expected skill set.

Given their dominance in Figure 4, these are not niche specialisations but the common technical language of the AI job market — a strong foundation across this shortlist likely matters more than depth in any single narrow tool.

3. Do not expect a remote-work premium or penalty.

Employers structuring offers, and candidates negotiating them, should not assume that a fully remote or fully on-site arrangement will shift the salary conversation in this market — the data shows no such relationship.

4. Factor in company size as a secondary, not primary, consideration.

Large companies do pay somewhat more at the median, but the effect is a fraction of the seniority effect — useful as a tie-breaker between similar offers, not as a primary job-search strategy.

5. Treat this dataset's findings as directional rather than definitive, given the structural pattern in Section 6.6.

The apparent synthetic banding between `years_experience` and `experience_level` means the seniority findings, while almost certainly directionally correct, would benefit from validation against a second, independently sourced dataset before being treated as a precise, real-world benchmark.

8.2 Technical Observations

`years_experience` was not included in the numeric correlation analysis.

Given its strong visual relationship with salary in Figure 12, adding it to `numeric_cols` in Section 3.4 would likely reveal the strongest numeric correlation in the whole dataset — a natural and low-effort extension of the existing analysis.

No boxplot of salary against `benefits_score` was produced.

The correlation matrix shows `benefits_score` has essentially no linear relationship with `salary`, but a dedicated boxplot or scatter plot (matching the treatment given to `remote_ratio` and `company_size`) would let this null result be confirmed visually and consistently, in the same style as every other categorical comparison in the report.

Some seaborn calls mix a fixed color with a hue variable.

In the violin plot (Figure 5), both `color="purple"` and `hue="salary_currency"` are specified together; seaborn resolves this by using the hue-based palette and effectively ignoring the fixed color argument. This did not cause an error and the chart renders correctly, but removing the redundant `color` argument in any hue-based plot would make the code's intent clearer.

The synthetic-looking experience banding (Section 2.5) is worth stating in any future publication of this analysis.

Flagging this pattern transparently, as this report does, is preferable to presenting the seniority findings as an unqualified real-world benchmark.

8.3 Directions for Further Work

7. Add `years_experience` to the correlation matrix and pairplot alongside `salary_usd`, `remote_ratio`, and `benefits_score`.
8. Produce a dedicated chart of `salary` against `benefits_score` to visually confirm the near-zero correlation found numerically.
9. Break down the top-skills analysis (Figure 4) by experience level, to test whether the specific skills in demand shift with seniority.
10. Cross-tabulate industry against `salary`, which was not examined in this notebook despite being an available column, as a likely additional source of categorical signal.
11. Given the strength and consistency of the seniority effect, this dataset would also be a strong candidate for a follow-up predictive-modelling project — mirroring the regression workflow used in the earlier student-performance and insurance-charges projects — to quantify precisely how much of salary variance experience level and job title can jointly explain.

9. Conclusion

This project set out to explore which factors are most strongly associated with AI job salaries, and which commonly assumed drivers turn out to matter less than expected.

The results are clear on both counts. **Seniority is, by a wide margin, the most consistent and substantial driver of salary** in this dataset, with average pay roughly tripling from Entry to Executive level across every chart that examines it. Required technical skills cluster around a clear, cross-disciplinary core — Python, SQL, cloud platforms, and ML frameworks — reflecting how thoroughly AI roles blend software engineering and data science.

Just as informative, however, are the factors that **do not** drive salary here. Formal education level shows almost no measurable premium — a PhD posting pays, on median, almost the same as a Bachelor's posting. Remote work arrangement shows no relationship with pay at all. And two available numeric columns, remote ratio and benefits score, both correlate with salary at essentially zero. Together these negative findings are as valuable as the positive one: they tell us plainly that whatever signal exists in this data lives in seniority and role structure, not in the handful of numeric attributes most readily available.

A necessary caveat runs through the whole analysis: the perfect, non-overlapping banding between `years_experience` and `experience_level` strongly suggests this dataset carries some synthetic or algorithmically-labelled structure, rather than being purely organic hiring data. This report has aimed to state that limitation plainly rather than let a clean-looking chart imply more real-world precision than the data can support — the seniority effect is almost certainly directionally real, but the exact banding boundaries should not be read as a literal industry standard.

Methodologically, this project also served a different purpose from the two regression projects that preceded it: it demonstrates that rigorous, well-documented exploratory analysis — correctly separating genuine signal from coincidental noise, and being honest about what a dataset's structure does and does not support — is a complete and valuable analytical exercise in its own right, independent of whether a predictive model is ultimately built on top of it.

10. References and Appendices

10.1 Dataset

AI Job Dataset (ai_job_dataset.csv)

A dataset of 15,000 AI-related job postings covering compensation, seniority, required skills, and employer characteristics, sourced from Kaggle.

10.2 Code Attribution

The base analytical code in this notebook — including the choice of charts, the skills-parsing logic, and the correlation setup — was adapted from a publicly available Kaggle notebook authored by **Omar Yasser**, with modifications made during adaptation for this report. The written analysis, interpretation, cross-referencing of findings across charts, and all recommendations in this document were researched and written independently by the author of this report.

[Note: the exact Kaggle notebook URL should be inserted here by the author, linking directly to Omar Yasser’s original notebook, to ensure complete and verifiable attribution.]

10.3 Library Documentation

Library	Documentation
pandas	https://pandas.pydata.org/docs/
NumPy	https://numpy.org/doc/
Matplotlib	https://matplotlib.org/stable/index.html
Seaborn	https://seaborn.pydata.org/

Specific API references:

- `seaborn.violinplot` — <https://seaborn.pydata.org/generated/seaborn.violinplot.html>
- `seaborn.pairplot` — <https://seaborn.pydata.org/generated/seaborn.pairplot.html>
- `pandas.to_datetime` — https://pandas.pydata.org/docs/reference/api/pandas.to_datetime.html
- `pandas.Series.value_counts` — https://pandas.pydata.org/docs/reference/api/pandas.Series.value_counts.html

Appendix A — Environment and Reproducibility

Item	Value
Language	Python 3
Notebook	aijobs.ipynb (Jupyter)
Source file	ai_job_dataset.csv
Records	15,000
Columns	19
Missing values	0
Duplicate rows	0
Modelling performed	None — exploratory/descriptive analysis only

Appendix B — Chart Index

Figure	Chart Type	Variables
1	Histogram + KDE	salary_usd
2	Bar chart	salary_currency (count)
3	Bar chart	experience_level vs. mean salary_usd
4	Horizontal bar chart	Top 20 required_skills (count)
5	Violin plot	experience_level vs. salary_usd, by salary_currency
6	Boxplot	experience_level vs. salary_usd
7	Boxplot	education_required vs. salary_usd
8	Boxplot	remote_ratio vs. salary_usd
9	Boxplot	company_size vs. salary_usd
10	Count plot	remote_ratio
11	Count plot	company_size
12	Scatter plot	years_experience vs. salary_usd, by experience_level
13	Bar chart	education_required (count)
14	Heatmap	salary_usd, remote_ratio, benefits_score (correlation)
15	Pairplot	salary_usd, remote_ratio, benefits_score

Appendix C — Correlation Values (Numeric Columns)

Variable Pair	Correlation
salary_usd – remote_ratio	+0.014
salary_usd – benefits_score	+0.001
remote_ratio – benefits_score	+0.003

Appendix D — Average Salary by Experience Level

Experience Level	Approx. Average Salary (USD)
EN (Entry)	\$63,000
MI (Mid)	\$88,000
SE (Senior)	\$122,000
EX (Executive)	\$188,000